Week 35: Footprint borders

Adult guide built from the problem files · Unpiloted. The card kit is physically unrehearsed.

Problems and routes

Problem	K–1	2–3	4–5	route
Slides that match and slides that do not	P1			first
Two different shortest slides	P2			first
A flip and a slide	P3			first
A longer flip-and-slide border	P4			first
Make the flip fail	P5			return
Half-turns	P6			return
Trade and test	P7			return
Two different shortest slides		P1		first
Two glides without their flips		P2		first
Shortest glide and shortest slide		P3		first
A flip without half-turns		P4		return
A half-turn without the flip		P5		return
Break a matching move		P6		return
Shortest slides and why			P1	first
Shortest slide and shortest glide			P2	first
What a 3 cm glide forces			P3	return
Two crossing flips			P4	return
Half-turns without flips			P5	return
Every matching move of one border			P6	return

First route. K–1 P1–P4; Grades 2–3 P1–P3; Grades 4–5 P1–P2. **Return visits.** K–1 P5–P7; Grades 2–3 P4–P6; Grades 4–5 P3–P6.

Key, one entry per problem

Slides that match and slides that do not

K–1 P1 · first route

Idea. A slide of a repeating border matches exactly when it carries every footprint to a footprint of the same kind and direction; a row spaced 4 cm apart matches after 4 cm and fails after 2 cm.

Two different shortest slides

K–1 P2 · first route

Idea. Different spacings give different shortest matching slides (4 cm and 6 cm rows of one footprint); a longer block made of two short repeats does not lengthen the shortest slide.

A flip and a slide

K–1 P3 · first route

Idea. A flip over the middle line followed by a slide along it (a glide) can match a border whose flip alone fails: footprints alternate between the rows, each half a period along.

A longer flip-and-slide border

K–1 P4 · first route

Idea. The same kind of match works at a longer spacing: with footprints 8 cm apart in each row, the flip and a 4 cm slide match where the 6 cm border used 3 cm, and the flip alone still fails.

Make the flip fail

K–1 P5 · return route

Idea. Pairing each footprint with its mirror across the middle line makes the flip match; moving or turning one kind of footprint in every repeat breaks it.

Half-turns

K–1 P6 · return route

Idea. A border with half-turned footprints alternating between the rows matches after a half-turn about a point between them; a border of same-direction footprints cannot match after any half-turn, because a half-turn reverses every footprint.

Trade and test

K–1 P7 · return route

Idea. Testing a partner's border with slides, flips, glides and half-turns checks every footprint, not just a few.

Two different shortest slides

Grades 2–3 P1 · first route

Idea. A repeating border has a shortest matching slide, its period; different spacings give different periods, and every matching slide is a whole number of periods.

Two glides without their flips

Grades 2–3 P2 · first route

Idea. A glide (flip over the middle line, then a nonzero slide along it) can match while the bare flip fails: upper footprints and their mirrors below alternate half a period apart, at any period (4 cm and 6 cm fit the boxes).

Shortest glide and shortest slide

Grades 2–3 P3 · first route

Idea. When the bare flip fails, the shortest matching glide is half the shortest matching slide, since two glides make a slide; two borders with different periods show the halving is a rule, not a coincidence.

A flip without half-turns

Grades 2–3 P4 · return route

Idea. A border with mirror pairs across the middle line matches the flip; no half-turn matches, because it would need half-turned or side-to-side mirrored footprints that are absent.

A half-turn without the flip

Grades 2–3 P5 · return route

Idea. Footprints and their half-turns alternating between the rows give half-turn centres on the middle line and no matching flip.

Break a matching move

Grades 2–3 P6 · return route

Idea. Moving footprints consistently in every repeat, possibly with a larger repeating part, can remove one symmetry while the border still repeats.

Shortest slides and why

Grades 4–5 P1 · first route

Idea. The shortest matching slide is the period P: a slide must take one footprint to the nearest footprint of the same kind and direction, so no shorter positive slide can match.

Shortest slide and shortest glide

Grades 4–5 P2 · first route

Idea. If a glide matches and the bare flip fails, glide displacements are odd multiples of P/2, so the shortest glide is P/2.

What a 3 cm glide forces

Grades 4–5 P3 · return route

Idea. A glide done twice is a slide of twice its length, so a matching 3 cm glide forces a matching 6 cm slide, but not a 3 cm slide (Border G is a counterexample).

Two crossing flips

Grades 4–5 P4 · return route

Idea. Flips over two perpendicular lines compose to a half-turn about their crossing; compositions of matching moves are matching moves.

Half-turns without flips

Grades 4–5 P5 · return route

Idea. Footprints and their half-turns alternating between the rows match half-turns about points on the middle line while every horizontal and vertical flip fails, since those would need mirrored footprints.

Every matching move of one border

Grades 4–5 P6 · return route

Idea. A complete symmetry description uses the generating rule and shows where every kind of footprint lands; sampling several moves is evidence, not proof.